

Prob 1 - Trigonometric series

a) $f(x) = \frac{1}{2} + \cos(2x) - 4\sin(4x)$

Just by looking at the function, we can see that we already have the Fourier series:

$$a_0 = \frac{1}{2}$$

$$a_n = \begin{cases} 1, & n=2 \\ 0, & \text{else} \end{cases}$$

$$b_n = \begin{cases} -4, & n=4 \\ 0, & \text{else} \end{cases}$$

b) $f(x) = 1 + \sin^2(x)$

We know that $\sin^2(x) = \frac{1}{2}(1 - \cos(2x)) \rightarrow$ rewriting the expression for $f(x)$.

$$f(x) = \frac{3}{2} - \frac{1}{2}\cos(2x), \quad P = \frac{2\pi}{2} = \pi \Rightarrow L = \frac{\pi}{2}$$

Just by looking at the function, we can see that we already have the Fourier series:

$$a_0 = \frac{3}{2}$$

$$a_n = \begin{cases} -\frac{1}{2}, & n=2 \\ 0, & \text{else} \end{cases}$$

$$b_n = 0$$

c) $f(x) = 1 - x, \quad -1 \leq x \leq 1, \quad T = 2L = 2 \Rightarrow L = 1$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx = \frac{1}{2} \int_{-1}^1 (1 - x) dx = \left[x - \frac{x^2}{2} \right]_{-1}^1 = 1$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = \int_{-1}^1 (1 - x) \cos(n\pi x) dx = \int_{-1}^1 \cos(n\pi x) dx - \int_{-1}^1 x \cos(n\pi x) dx = 2 \int_0^1 \cos(n\pi x) dx = \frac{2}{n\pi} [\sin(n\pi x)]_0^1 = 0$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = \int_{-1}^1 (1 - x) \sin(n\pi x) dx = \int_{-1}^1 \sin(n\pi x) dx - \int_{-1}^1 x \sin(n\pi x) dx = -2 \int_0^1 x \sin(n\pi x) dx$$

$$\rightarrow -2 \int_0^1 x \sin(n\pi x) dx = -2 \left[-\frac{x \cos(n\pi x)}{n\pi} - \int \frac{\cos(n\pi x)}{n\pi} dx \right]_0^1 = \frac{2}{n\pi} \left[x \cos(n\pi x) + \frac{\sin(n\pi x)}{n\pi} \right]_0^1 = \frac{2 \cos(n\pi)}{n\pi} = \frac{2(-1)^n}{n\pi}$$

d) $f(x) = |\cos(3\pi x)|, \quad T = 2L = \frac{2}{3} \Rightarrow L = \frac{1}{3}$

We see that $\cos(3\pi x)$ is always pos on int $x \in [-\frac{1}{6}, \frac{1}{6}]$. Don't need to worry about absv.

$$a_0 = \frac{1}{2L} \int_{-L}^L \cos(3\pi x) dx = 3 \cdot 2 \int_0^{\frac{1}{6}} \cos(3\pi x) dx = \frac{6}{\pi} [\sin(3\pi x)]_0^{\frac{1}{6}} = \frac{2}{\pi} \sin\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$$

$$a_n = \frac{1}{L} \int_{-L}^L \cos(3\pi x) \cos\left(\frac{n\pi x}{L}\right) dx = 6 \int_{-\frac{1}{6}}^{\frac{1}{6}} \cos(3\pi x) \cos(n\pi x) dx = 3 \int_{-\frac{1}{6}}^{\frac{1}{6}} \cos(3\pi x + n\pi x) + \cos(3\pi x - n\pi x) dx$$

$$\rightarrow 6 \int_0^{\frac{1}{6}} \cos(3\pi x(1+n)) dx + 6 \int_0^{\frac{1}{6}} \cos(3\pi x(1-n)) dx = \frac{6}{3\pi(1+n)} [\sin(3\pi x(1+n))]_0^{\frac{1}{6}} + \frac{6}{3\pi(1-n)} [\sin(3\pi x(1-n))]_0^{\frac{1}{6}}$$

$$= \frac{6}{3\pi(1+2n)} \sin\left(\frac{\pi}{2}(1+2n)\right) + \frac{6}{3\pi(1-2n)} \sin\left(\frac{\pi}{2}(1-2n)\right)$$

$$= \frac{2(-1)^n}{\pi(1+2n)} + \frac{2(-1)^n}{\pi(1-2n)} = \frac{4(-1)^n}{\pi} \left(\frac{1}{1+2n} + \frac{1}{1-2n} \right) = \frac{4(-1)^n}{\pi(1-4n^2)}$$

$$b_n = \frac{1}{1/6} \int_{-1/6}^{1/6} \cos(3\pi x) \sin\left(\frac{n\pi x}{1/6}\right) dx = 0$$

even · odd ⇒ odd = 0

Prob 2 - Complex series

a) $F(x) = -3e^{-4ix} + e^{-ix} + 4e^{3ix}$

$$e^{ix} = \cos(x) + i\sin(x) \Rightarrow e^{-4ix} = \cos(4x) - i\sin(4x) \Rightarrow T=2L=\frac{\pi}{2}$$

$$\Rightarrow e^{-ix} = \cos(x) + i\sin(x) \Rightarrow T=2L=2\pi \quad \text{Overall period} \Rightarrow L=\pi$$

$$\Rightarrow e^{3ix} = \cos(3x) + i\sin(3x) \Rightarrow T=2L=\frac{2\pi}{3}$$

$$F(x) \sim \sum c_k e^{\frac{ik\pi x}{L}} = \sum c_k e^{ikx} \Rightarrow c_k = \begin{cases} -3, & n=-4 \\ 1, & n=-1 \\ 4, & n=3 \end{cases}$$

b) $F(x) = \cos^2(x), \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}, \quad T=2L=\pi \Rightarrow L=\frac{\pi}{2}$

$$\cos^2(x) = \frac{1 + \cos(2x)}{2}, \quad \cos(2x) = \frac{e^{2ix} + e^{-2ix}}{2} \Rightarrow F(x) = \frac{1}{2} + \frac{e^{2ix}}{4} + \frac{e^{-2ix}}{4}$$

$$c_0 = a_0 = \frac{1}{2L} \int_{-L}^L F(x) dx = \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1 + \cos(2x)}{2} dx = \frac{1}{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 + \cos(2x) dx = \frac{1}{2\pi} \left[x + \frac{\sin(2x)}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{1}{2}$$

$$F(x) \sim \sum c_k e^{\frac{ik\pi x}{L}} = \sum c_k e^{2ikx} = \frac{1}{2} + \frac{e^{2ix}}{4} + \frac{e^{-2ix}}{4} \Rightarrow c_k = \begin{cases} 1/4, & n=-1 \\ 1/2, & n=0 \\ 1/4, & n=1 \end{cases}$$

c) $F(x) = x^2 - x, \quad -1 \leq x \leq 1, \quad T=2L=2 \Rightarrow L=1$

$$c_0 = a_0 = \frac{1}{2L} \int_{-L}^L F(x) dx = \frac{1}{2} \int_{-1}^1 x^2 - x dx = \int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

$$c_n = \frac{1}{2L} \int_{-L}^L F(x) e^{-in\pi x} dx = \frac{1}{2} \int_{-1}^1 (x^2 - x) e^{-in\pi x} dx = \frac{1}{2} \left(\int_{-1}^1 x^2 e^{-in\pi x} dx - \int_{-1}^1 x e^{-in\pi x} dx \right)$$

$$\int_{-1}^1 x e^{-in\pi x} dx = \left[-\frac{x e^{-in\pi x}}{in\pi} - \int \frac{e^{-in\pi x}}{in\pi} dx \right]_{-1}^1 = \left[-\frac{x e^{-in\pi x}}{in\pi} + \frac{e^{-in\pi x}}{n^2\pi^2} \right]_{-1}^1 = \left(-\frac{e^{-in\pi}}{in\pi} + \frac{e^{-in\pi}}{n^2\pi^2} \right) - \left(\frac{e^{in\pi}}{in\pi} + \frac{e^{in\pi}}{n^2\pi^2} \right) = -\frac{1}{in\pi} (e^{-in\pi} + e^{in\pi}) + \frac{1}{n^2\pi^2} (e^{-in\pi} - e^{in\pi}) = \frac{1}{n^2\pi^2} - \frac{1}{in\pi} (\cos(n\pi)) = (-1)^n$$

$$\int_{-1}^1 x^2 e^{-in\pi x} dx = \left[-\frac{x^2 e^{-in\pi x}}{in\pi} - 2 \int x e^{-in\pi x} dx \right]_{-1}^1 = \left[-\frac{x^2 e^{-in\pi x}}{in\pi} - 2 \left(-\frac{x e^{-in\pi x}}{in\pi} + \frac{e^{-in\pi x}}{n^2\pi^2} \right) \right]_{-1}^1$$

$$= \left[-\frac{e^{-in\pi}}{in\pi} - 2 \left(-\frac{e^{-in\pi}}{in\pi} + \frac{e^{-in\pi}}{n^2\pi^2} \right) \right] - \left[-\frac{e^{in\pi}}{in\pi} - 2 \left(-\frac{e^{in\pi}}{in\pi} + \frac{e^{in\pi}}{n^2\pi^2} \right) \right]$$

d) $F(x) = e^{-x}, \quad -1 \leq x \leq 1, \quad T=2L=2 \Rightarrow L=1$

$$c_0 = a_0 = \frac{1}{2L} \int_{-L}^L F(x) dx = \frac{1}{2} \int_{-1}^1 e^{-x} dx = -\frac{1}{2} [e^{-x}]_{-1}^1 = -\frac{1}{2} \left(\frac{1}{e} - e \right) = \frac{e}{2} - \frac{1}{2e} = \frac{e^2 - 1}{2e}$$

$$c_n = \frac{1}{2L} \int_{-L}^L F(x) e^{-in\pi x} dx = \frac{1}{2} \int_{-1}^1 e^{-x} e^{-in\pi x} dx = \frac{1}{2} \int_{-1}^1 e^{-x(1+i\pi n)} dx = -\frac{1}{2(1+i\pi n)} [e^{-x(1+i\pi n)}]_{-1}^1 = \frac{e^{-1-i\pi n} - e^{1+i\pi n}}{2(1+i\pi n)} = \frac{\sinh(1) - (-1)^n \cosh(1)}{1+i\pi n}$$

Prob 3 - Parseval's theorem

a) The formula for E_N is given by

$$E_N = \int_{-L}^L (f(x) - F_N(x))^2 dx = \int_{-L}^L f^2(x) dx - L \sum_{k=-N}^N |c_k|^2 / (2a_0^2 + \sum a_n^2 + b_n^2)$$

b)

$$E_N = \int_{-1}^1 (e^{-x})^2 dx - 2 \sum_{k=-N}^N |c_k|^2$$

$$\int_{-1}^1 (e^{-x})^2 dx = \int_{-1}^1 e^{-2x} dx = -\frac{1}{2} [e^{-2x}]_{-1}^1 = \frac{e^2 - e^{-2}}{2} = \sinh(2)$$

$$E_N = \sinh(2) - 2 \sum_{k=-N}^N \left| \frac{\sinh(1)}{1 + ik\pi} (-1)^k \right|^2 = \sinh(2) - 2 \sum_{k=-N}^N \frac{\sinh^2(2)}{(1 + k^2 \pi^2)}$$

c)

$$g(x) = e^{-|x|}, \quad -1 \leq x \leq 1, \quad T = 2h = 2 \Rightarrow h = 1$$

$$c_n = \frac{1}{2h} \int_{-L}^L g(x) e^{-inxh} dx$$

$$= \frac{1}{2} \int_{-1}^1 e^{-|x|} e^{-inx} dx$$

$$= \frac{1}{2} \int_{-1}^0 e^x e^{-inx} dx + \frac{1}{2} \int_0^1 e^{-x} e^{-inx} dx$$

$$= \frac{1}{2} \int_{-1}^0 e^{x(1-in)} dx + \frac{1}{2} \int_0^1 e^{-x(1+in)} dx$$

$$= \frac{1}{2} \left[\frac{e^{x(1-in)}}{1-in} \right]_{-1}^0 + \frac{1}{2} \left[\frac{e^{-x(1+in)}}{-(1+in)} \right]_0^1$$

$$= \frac{1}{2} \left(\frac{e^{1-in}}{1-in} - \frac{1}{1-in} \right) + \frac{1}{2} \left(\frac{e^{-1-in}}{-(1+in)} - \frac{1}{-(1+in)} \right)$$

$$= \frac{e^{1-in}}{2(1-in)} - \frac{1}{2(1-in)} + \frac{e^{-1-in}}{2(1+in)} - \frac{1}{2(1+in)}$$

=

Prob 4 - Gibbs phenomenon

$$f(x) = 1 - x^2, \quad x \in [0, 1]$$

We already know that $f(x)$ is odd $\Rightarrow a_0 = a_n = 0$. Still need to compute b_n

$$b_n = \frac{2}{1} \int_0^1 (1-x^2) \sin(n\pi x) dx$$

$$= 2 \int_0^1 \sin(n\pi x) dx - 2 \int_0^1 x^2 \sin(n\pi x) dx$$

$$= 2 \left[-\frac{\cos(n\pi x)}{n\pi} \right]_0^1 + 2 \left[\frac{x^2 \cos(n\pi x)}{n\pi} + \frac{2}{n\pi} \int x \cos(n\pi x) dx \right]_0^1$$

$$= \frac{-2}{n\pi} (\cos(n\pi) - 1) + \frac{2}{n\pi} \left[x^2 \cos(n\pi x) + 2 \left(\frac{x \sin(n\pi x)}{n\pi} - \frac{1}{n\pi} \int \sin(n\pi x) dx \right) \right]_0^1$$

$$= \frac{2(1-(-1)^n)}{n\pi} + \frac{2(-1)^n}{n\pi} + \frac{4}{n^2\pi^2} \left[x \sin(n\pi x) + \frac{\cos(n\pi x)}{n\pi} \right]_0^1$$

$$= \frac{2}{n\pi} + \frac{4}{n^2\pi^2} \left(\frac{\cos(n\pi)}{n\pi} - 1 \right)$$

$$= \frac{2}{n\pi} \left(1 + \frac{2(-1)^n - 2}{n^2\pi^2} \right)$$

